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1. Introduction

Autonomous data collection and ingestion pipelines lie at the heart of many modern artificial intelligence systems (Steidl, Felderer and Ramler, 2023). As organisations and researchers increasingly rely on large-scale automated acquisition of data for training, monitoring, or inference tasks, the design and robustness of those pipelines become critical. One emerging risk, seldom addressed in structured research, is the phenomenon of gradual degradation, inefficiency, bias reinforcement, redundancy accumulation, and resource escalation, here referred to as “AI Crawler Tar Pits”. The metaphor emphasises how even intelligent, adaptive retrieval systems can become stuck in self-reinforcing inefficiency, reminiscent of software development “tar pits”.

To investigate these risks concretely, this research adapts the idea of “crawler tar pits” to the domain of network traffic ingestion and analysis, by using the “CTU-13 Dataset”. We propose a new working dataset designation, CTU-13 Dataset, tailored to study how data ingestion pipelines, similar in structure to web-crawlers or autonomous agents, may degrade when repeatedly ingesting large volumes of network traffic and applying machine-learning-based filtering or classification.

The “CTU-13” dataset is a collection of real-world network traffic traces created in thirteen situations at the “Czech Technical University (CTU)” and consisting of botnet traffic, normal traffic, and background traffic (*The CTU-13 Dataset. A Labeled Dataset with Botnet, Normal and Background traffic.*, 2025). The various scenarios are associated with the various malware or botnet family, and records of those sceneries have been extensively utilised in botnet detecting studies. Its realism, variety, and naming make it a potent base to simulate the ingestion and classification pipes that may be like AI crawlers. This study seeks to identify structural and functional reasons resulting in poor ingestion quality using CTU-13-TarPitNet and is reminiscent of the dynamics seen in the web-scale crawling of tar-pits.

This research is timely and relevant. As cyber-security, intrusion detection, and anomaly detection systems increasingly incorporate automated data pipelines and self-training models, the risk that such systems may gradually degrade or entrench bias becomes significant. Understanding “tar pit” phenomena in this context can guide the design of more robust, efficient, and maintainable data-driven AI systems.

This proposal details a plan to deliver a conceptual framework, empirical evidence, and practical mitigation strategies rooted in real-world network traffic data.

2. Research Aim & Objectives

2.1 Aim

- To investigate and characterise the emergence of “tar-pit” phenomena in long-term automated data-ingestion and classification pipelines, using the **CTU-13-TarPitNet** dataset as a concrete case study, and to propose and evaluate design strategies that mitigate redundancy, data drift and performance degradation.

2.2 Objectives

- To implement and simulate multiple ingestion-classification pipeline configurations on CTU-13-Daatset, mimicking real-world autonomous data ingestion.
- To perform repeated ingestion cycles, including normal, perturbed, and drift-injected data batches to observe how pipeline configurations behave over time under evolving data distributions.
- To statistically compare the performance of different models with respect to both classification effectiveness (precision, recall, F1-score, AUC) and long-term efficiency/diversity metrics.

3. Research Questions

- Under what conditions do automate ingestion-classification pipelines begin to exhibit tar-pit phenomena (e.g., increasing redundancy, declining novelty, resource inefficiency) when processing network-traffic data?
- How do different pipeline configurations, in particular deduplication filtering, diversity-aware sampling, and periodic retraining, influence the onset, severity, and trajectory of tar-pit conditions?
- What trade-offs arise between classification performance (detection of malicious flows) and long-term pipeline health (data diversity, resource usage, redundancy)?
- How does the injection of data drift or scenario perturbation (e.g., new botnet families, altered traffic patterns) affect pipeline stability and tar-pit indicators under different configurations?

4. Literature Review

This literature review explores prior studies related to network-traffic datasets, machine learning for botnet detection, challenges in automated data ingestion, and conceptual parallels between crawler inefficiency and long-running AI data pipelines. The objective is to synthesise existing knowledge, identify relevant limitations, and delineate gaps that justify the notion of “AI Crawler Tar Pits” applied to network traffic ingestion.

4.1 The CTU-13 Dataset and Its Use in Botnet Detection

The CTU-13 Dataset originates from a controlled study at CTU University, Czech Republic, designed to capture realistic botnet behaviour associated with multiple malware families. Each of the thirteen captures (scenarios) corresponds to a different botnet seed. Traffic records

include botnet flows injected into otherwise benign or background traffic, reflecting real-world network environments (*The CTU-13 Dataset. A Labeled Dataset with Botnet, Normal and Background traffic.*, 2025)

Researchers have widely adopted CTU-13 for developing and evaluating botnet detection methods. For instance, studies using decision tree, random forest, and neural network classifiers report reasonably high performance for distinguishing botnet vs normal traffic (Padhiar and Patel, 2023). In addition, graph-based clustering approaches have been proposed to detect botnets by transforming netflow features into graph representations, aiming to capture communication patterns instead of relying solely on flow-level statistics (Chowdhury *et al.*, 2017)

A recent review of CTU-13 dataset usage suggests that despite its age (2011 capture), it remains relevant because it offers real botnet traces, multi-botnet diversity, and labelled flows with a mixture of benign and malicious traffic (Vishwakarma, 2020). These properties make CTU-13 a strong candidate for research on automated ingestion and classification pipelines. (Chowdhury *et al.*, 2017); (Issac, Fryer and Jacob, 2024). The class imbalance, mixed data types (categorical and numerical), and drift across scenarios complicate modelling and generalisation.

Furthermore, most existing work treats CTU-13 as a static dataset: each model is trained and evaluated once, without exploring the dynamics of repeated ingestion, retraining, or continuous data updates. This leaves unexamined the long-term stability and potential degradation of systems ingesting data over time, a central concern for this proposal’s “tar pit” perspective.

4.2 Machine Learning-based and Anomaly Detection Approaches on CTU-13

Beyond traditional supervised classifiers (e.g., decision trees, random forests), more advanced methods have been proposed to leverage sequential and temporal aspects of network traffic. For instance, a study introduced a recurrent variational autoencoder (RVAE) trained on network flow sequences to detect botnet traffic in streaming mode. This method demonstrated detection performance comparable to state-of-the-art baselines when evaluated on CTU-13 (Kim *et al.*, 2020). The RVAE approach illustrates how deep learning can model complex temporal dependencies, offering potential advantages over static flow-based classification.

Nevertheless, even such advanced methods inherit some limitations. Since they rely on prior data distributions for normal behaviour, they may become vulnerable when traffic patterns evolve. Repeated ingestion and retraining cycles with similar data may encourage overfitting to prevalent patterns, especially if the ingestion pipeline lacks mechanisms to ensure data diversity or resist redundancy. These issues resonate strongly with the tar pit metaphor, where repeated cycles of ingestion and filtering gradually degrade the system’s ability to extract genuinely novel or useful information.

Moreover, a broader criticism in intrusion detection research concerns the quality and representativeness of publicly available datasets. A study on dataset generation for intrusion detection (ID2T) highlighted that many existing datasets suffer from negative qualities, such as lack of background traffic realism, temporal correlation, or diversity, which compromises

their usefulness for benchmarking (Cordero *et al.*, 2019). The authors propose injecting synthetic attacks into realistic background traffic to produce more representative datasets. This suggests that CTU-13, while valuable, may not fully reflect the diversity and unpredictability of modern network environments, especially under continuous ingestion or evolving attack patterns.

4.3 Dataset Degradation, Drift, Redundancy and Self-Reinforcing Loops

Outside the domain of cybersecurity, the broader literature on machine learning pipelines warns against data drift, dataset contamination, and self-reinforcing loops when models are retrained on their own outputs or on data collected under their influence. Studies in general AI and data science have shown that automated pipelines lacking human oversight can gradually reinforce bias, reduce data diversity, and accumulate duplicates or low-quality samples. In classification systems, small upstream errors may cascade during retraining, producing compounding degradation over time (uplatzblog, 2025); (Rockett, 2025)

In the context of network traffic ingestion, similar risks exist, i.e., if ingestion and filtering decisions depend heavily on ML classifiers or heuristics, then repeated ingestion can reinforce prior biases, cause data homogeneity, and exclude rare but potentially important anomalies. This feedback loop can result in diminishing returns: growing data volume without commensurate increase in useful, novel information.

4.4 The “Tar Pit Analogy” and Its Relevance to Automated Data Ingestion

The concept of a “tar pit” in software engineering refers to the phenomenon where software projects become mired in complexity, maintenance burden, and inefficiency, despite continued effort (Marks, 2006); (*Into the tar pit*, 2020). While this metaphor originated in the context of software development, it can be adapted to describe data ingestion pipelines. In this adapted sense, an “AI Crawler Tar Pit” refers to an automated ingestion system that, over time, becomes trapped in inefficiency, redundancy, data drift, bias amplification, and resource waste.

Although literature explicitly using the term “crawler tar pit” is sparse, many of the underlying components have been studied. The challenges of dataset degradation, self-reinforcing loops, model drift, dataset bias, and redundancy have been addressed in various contexts, but seldom unified under a single conceptual framework. The absence of such a framework suggests a gap where this research can contribute.

While CTU-13 is extensively used in botnet detection, existing journal studies analyze it as static captures without evaluating long-term ingestion pipelines under repeated cycles or metrics for degradation, redundancy, and inefficiency (Xing *et al.*, 2021); (Ring *et al.*, 2019). This gap limits understanding of how automated pipelines behave in production-like continuous ingestion scenarios.

4.5 Related Work Beyond CTU-13: Recent Dataset Advances and Limitations

Recent work in network intrusion detection has acknowledged some of these limitations. For instance, a dataset published in 2023, TII-SSRC-23 Dataset, was introduced to provide more diverse traffic patterns, updated to reflect contemporary network behaviour, and to improve representativeness beyond older datasets (Herzalla, Lunardi and Lopez, 2023). Such datasets

attempt to address temporal changes, diversity, and modern traffic patterns. However, their use for long-term ingestion pipeline evaluation remains uncommon.

Complementary work on adversarial dataset augmentation, such as studies of datasets with obfuscated network traffic or evasive behaviour, demonstrate that classifiers trained on clean datasets (like CTU-13) often fail to generalise under adversarial conditions (Herzalla, Lunardi and Lopez, 2023). These results further emphasise the necessity of robust ingestion pipelines that can resist degradation and maintain detection quality under evolving or adversarial traffic.

However, these works focus primarily on detection performance rather than on understanding pipeline-level efficiency, redundancy, or long-term stability. Thus, the broader issue of tar-pit phenomena in ingestion pipelines remains underexplored.

4.6 Summary of Gaps and Justification for This Research

In summary, the literature offers rich insight into botnet detection using CTU-13, the design of ML-based detection systems, challenges in dataset representativeness, and the risks of data drift or redundancy in automated pipelines. Nevertheless, several gaps remain:

- Most prior work treats datasets as static and does not simulate repeated ingestion or continuous pipeline operation.
- There is no unified framework defining or measuring tar-pit phenomena (redundancy accumulation, drift, computational inefficiency, feedback loops) in data ingestion systems.
- Mitigation strategies (e.g., data augmentation, diversity sampling, adversarial augmentation) have been proposed, but seldom in the context of long-term ingestion stability and resource efficiency.
- Newer datasets attempt to improve diversity and traffic realism, yet there is little exploration of how ingestion pipelines process them over time or under continuous deployment.

Given these gaps, applying the “AI Crawler Tar Pits” concept to network traffic ingestion, using CTU-13-TarPitNet Dataset, is both novel and necessary. This research can provide conceptual clarity, empirical evidence, and practical guidelines for designing resilient ingestion pipelines for cybersecurity and other AI systems.

5. Research Methodology

This section describes the planned methodological approach to investigate tar-pit phenomena in network traffic ingestion pipelines using the CTU-13-TarPitNet Dataset. The study involves data preparation, pipeline construction, simulation of ingestion cycles, measurement of degradation, and evaluation of mitigation strategies.

5.1 Data Preparation and Preprocessing

The base data will be drawn from the CTU-13 Dataset, accessed via the publicly available repository hosting packet captures and netflow files (*The CTU-13 Dataset. A Labeled Dataset*

with Botnet, Normal and Background traffic., 2025). Prior to ingestion experiments, netflow files will be converted to a standard tabular format suitable for machine learning pipelines. Each record includes features such as source address, destination address, protocol, source port, destination port, start time, duration, byte counts, packet counts, direction, type-of-service fields, and label (botnet, normal, background) as defined in CTU-13.

To prepare the dataset for repeated ingestion simulations, flows from several scenarios will be aggregated to create a large base corpus. Additional synthetic transformations will be applied to simulate temporal drift, variation in traffic patterns, and injection of noise or duplicates. For example, variations may include reassigning timestamps, shuffling flow ordering, re-encoding categorical fields, or duplicating flows with minor perturbations. This extended dataset will approximate real-world ingestion conditions where traffic recurs over time, changes structure or arrives from multiple sources.

5.2 Pipeline Architecture

An ingestion pipeline analogous to a crawler will be implemented. The pipeline comprises the following modules:

- **Loader Module**
 - Reads netflow or tabular data and standardises format.
- **Feature Extraction Module**
 - This module transforms raw flows into model-ready feature vectors, including one-hot encoding of categorical data, scaling of numerical features, and optional feature engineering (e.g., aggregating flows per time window, protocol distribution vectors, traffic burst metrics).
- **Classifier module**
 - A machine-learning model (e.g., decision tree, random forest, or neural network) trained to label flows as botnet or benign/background. A baseline classifier (e.g., random forest) will be used to reflect typical ingestion pipelines in intrusion detection research.
- **Feedback Loop Module**
 - After classification, classified flows are stored; for repeated ingestion cycles, the stored flows influence future ingestion or filtering decisions (for example, flows like previously accepted ones may be downweighted). This simulates real-world pipelines that filter or prioritise based on previously ingested data.
- **Storage Module**
 - Database or file-based storage that keeps raw flows, metadata (ingestion timestamp, classification label, feature hash), and provenance information.

5.3 Evaluation Methods

Because the target dataset (CTU-13-TarPitNet Dataset) is classification-based (botnet vs benign/background network flows), the evaluation strategy must rely on classification-appropriate metrics. The following metrics and indicators will be used to assess both classification performance and long-term ingestion pipeline behaviour, with emphasis on detecting tar-pit phenomena.

- **Confusion Matrix**
 - For each ingestion cycle (or evaluation batch), compute the standard confusion matrix summarising true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). This serves as the foundation for all downstream performance metrics (Albanhawry, Khalil and Moustafa, 2024)
- **Accuracy**
 - The proportion of correctly classified flows (both positive and negative) over all predictions. Calculated as $(TP + TN) / (TP + TN + FP + FN)$. Useful as a coarse overall indicator, though of limited reliability if class imbalance exists
- **Precision (Positive Predictive Value)**
 - $TP / (TP + FP)$. Indicates how many of the predicted “botnet” flows are “actually” malicious; important when false positives are costly.
- **Recall (Sensitivity / True Positive Rate)**
 - $TP / (TP + FN)$. Measures the ability to detect actual malicious flows; relevant when false negatives (missed botnets) are especially undesirable.
- **F1-Score**
 - Harmonic mean of precision and recall: $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$. Favoured when there is class imbalance or when a balance between precision and recall is required.
- **ROC–AUC (Area Under the Receiver Operating Characteristic Curve)**
 - For classifiers that assign continuous or probabilistic scores (rather than binary labels), use ROC curves to evaluate the trade-off between true positive rate and false positive rate across thresholds. The area under this curve provides a threshold-independent measure of discrimination. This metric is widely used in intrusion detection and network classification contexts

5.4 Ethical Considerations and Reproducibility

Since CTU-13 is publicly available and anonymised to avoid user privacy issues, using it poses no privacy risk. All data use will comply with dataset licensing and attribution requirements. The codebase of the ingestion pipeline, along with configuration scripts for experiments, will be released under an open-source licence to ensure reproducibility. Documentation will include detailed descriptions of preprocessing, pipeline design, sampling policies, and metric computation.

6. Expected Research Outcomes

This research is expected to yield multiple contributions to both theory and practice. First, it will formalise the concept of AI Crawler Tar Pits in the context of network traffic ingestion, providing a taxonomy of degradation phenomena (redundancy accumulation, novelty collapse, resource inefficiency, classification drift) and a set of quantitative indicators to identify tar-pit states.

Second, the experiments using CTU-13-TarPitNet Dataset will generate empirical evidence demonstrating whether and how ingestion pipelines degrade over repeated cycles, under various pipeline configurations, and in presence of drift or noise. This empirical grounding will highlight real risks in long-term autonomous ingestion systems; a topic seldom explored in current literature.

Third, the project will deliver practical mitigation strategies: pipeline designs (diversity-aware sampling, duplication detection, adaptive retraining), parameterisation guidance (e.g., similarity thresholds, batch sizes, retraining intervals), and policy recommendations for organisations deploying continuous ingestion systems.

Fourth, the code and experiment suite will be released publicly, allowing other researchers to reproduce, extend, or adapt the framework for other datasets or domains.

Finally, the research may generalise beyond cybersecurity: the conceptual framework and methodology could apply to any data-driven system ingesting large-scale streams (web crawling, log ingestion, social media scraping, sensor networks). Thus, the results could inform best practices for sustainable, robust, and efficient data pipelines across disciplines.

7. Proposed Project Timeline

The research will proceed over twelve months according to the following schedule:

Table 1: Proposed Project Timeline

Month	Activities / Milestones
Month 1	Finalising dataset preparation plan; download CTU-13 data; design preprocessing pipeline; define feature schema and ingestion architecture. Preparation of extended ingestion dataset: aggregate flows; apply synthetic transformations; generate noise and perturbations to simulate drift.

Month 2	Implementation of ingestion pipeline modules: loader, feature extraction, storage system, base classifier, duplication detection, sampling and filtering modules. Implementing different ingestion configurations (baseline, deduplication-based, diversity-aware sampling, adaptive retraining). Defining and implement metric computation modules (redundancy, novelty, resource cost per useful flow, classification performance, storage vs utility).
Month 3	Running ingestion simulations across all pipeline configurations; collect time-series data; compute metrics; log resource usage. Introducing drift and noise scenarios; executing stress-testing ingestion runs; retrain pipelines where applicable; monitor performance, redundancy, novelty, and classification stability.
Month 4	Performing statistical analysis and comparative evaluation across configurations; conduct ablation and parameter-tuning studies (similarity thresholds, sampling strategies, retraining intervals, batch sizes). Consolidation of results; preparation of documentation of findings; draft conceptual framework and mitigation guidelines. Finalising thesis; releasing code and dataset scripts; preparation of final report for submission.

8. Conclusion

Autonomous data ingestion systems are increasingly central to AI-driven domains including cybersecurity, analytics, and large-scale monitoring. Yet the risk that such systems may gradually degrade, failing to deliver novel or useful data, consuming disproportionate resources, reinforcing bias, or collapsing into redundancy, remains under-explored. By applying the concept of AI Crawler Tar Pits to network-traffic ingestion and using the CTU-13-TarPitNet Dataset as a testbed, this research aims to illuminate these risks with empirical evidence, rigorous measurement, and practical mitigation strategies. The anticipated theoretical framework, diagnostic metrics, ingestion policies, and open-source pipeline will provide a foundation for designing more resilient, efficient, and sustainable data infrastructures. The insights will benefit not only cybersecurity applications but any discipline relying on continuous, automated data collection. In doing so, the study contributes to safer and more dependable AI pipelines in the long term.

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